Pritesh Rodge

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Education

Indian Institute of Information Technology, Design and Manufacturing Jabalpur, M.P
Bachelors of Technology in Computer Science and Engineering *Aug. 2023 - present*

Experience

Robotics Software Intern - UGO Inc. Jun 2026 - July 2026
Summer Intern 2026 *Tokyo, Japan - Hybrid*

- Engineered an edge-based digital meter reading pipeline using **YOLOv8n** and **PaddleOCR**, leveraging hardware **NPU optimization** to achieve a **43x speedup** and reduce total inference time to **under 50ms**.
- Overhauled a modular analog meter reading architecture by replacing heavy models and **migrating model executions to NPU**, resulting in a **97.38% reduction in execution time from over 304 seconds to under 8 seconds**.
- Resolved severe 4K camera detection failures at a 1-meter distance by implementing systematic detect-segment-recognition approach, achieving **over 96% final reading accuracy** across both digital and analog systems.
- Developed an interactive Quick-Train application with a guided labeling interface, enabling on-site users to seamlessly capture 30 to 40 image datasets and **auto-train models** to recognize **unseen meter types**.

Linux Kernel Mentorship (LFX) – The Linux Foundation June 2025 - August 2025
Linux Kernel Bug Fixing Summer Mentorship 2025 *Remote*

- Debugged **Linux kernel crashes** by reproducing Syzbot-reported issues using **Syzkaller, QEMU, and GDB**.
- Traced **memory corruption and overflow bugs** through low-level code analysis and **runtime debugging**.
- Developed and submitted **kernel patches** via mailing lists, incorporating reviewer feedback to meet **upstream standards**.
- Gained deep understanding of **kernel internals** including **system calls, scheduling, and memory management** through hands-on debugging.

Projects

Autonomous Warehouse Drone – eYRC 2024-25
ROS 2, Gazebo Fortress, Python, C++, PID Control, Path Planning, Image Processing

- Built an autonomous nano-quadcopter system to navigate warehouse aisles, detect packages, and stabilize precisely at target setpoints in a dynamic environment.
- Implemented a custom PID controller achieving stabilization within an error margin of ±0.4, enabling accurate waypoint navigation and drone control Video
- Used ROS 2 nodes with Gazebo simulations for localization via WhyCon markers, warehouse mapping, and efficient path-planning across aisles Video

Technical Skills

- **Languages:** Python, SQL, C/C++, Java.
- **Libraries & Frameworks:** PyTorch, Pandas, NumPy, scikit-learn, Matplotlib, FastAPI.
- **Systems & Tools:** Linux, Git, Docker, QEMU, GDB, ROS 2, Gazebo.
- **Core Concepts:** Machine Learning, Data Pipelines, Data Structures, Operating Systems, DBMS, OOPS.

Skills/Leadership/Volunteering

- Solved **400+** problems across **LeetCode**, **CodeChef**, and **CodeForces**, strengthening problem-solving across data structures, graphs, dynamic programming, trees, and strings.
- Ranked Top **25** among **700+** teams nationally in the **e-Yantra Robotics Competition (eYRC 2024-25)** through development of an autonomous warehouse drone system.
- Led robotics system development for national competitions including **Robocon 2025**, coordinating design, testing, and integration across sub-teams.
- Mentored **300+** school students through hands-on robotics workshops, simplifying complex technical concepts into practical demonstrations.
- Coordinated inter-college events as part of the **IIITDMJ Football Club**, managing logistics, scheduling, and team collaboration under constraints.